

CSC3209 Software architecture and Design Patterns

Lecture 7: Understanding Quality Attributes More Details and Scenarios

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References

- Materials from
 - Chapter 5,6,7,8,9,10,11: Len Bass, Paul Clements, Rick Kazman, Software Architecture in Practice, 3/E

Lecture Outline

- Availability
- Interoperability
- Modifiability
- Performance
- Security
- Testability
- Usability

Availability

- What is Availability?
- Availability General Scenario

What is Availability?

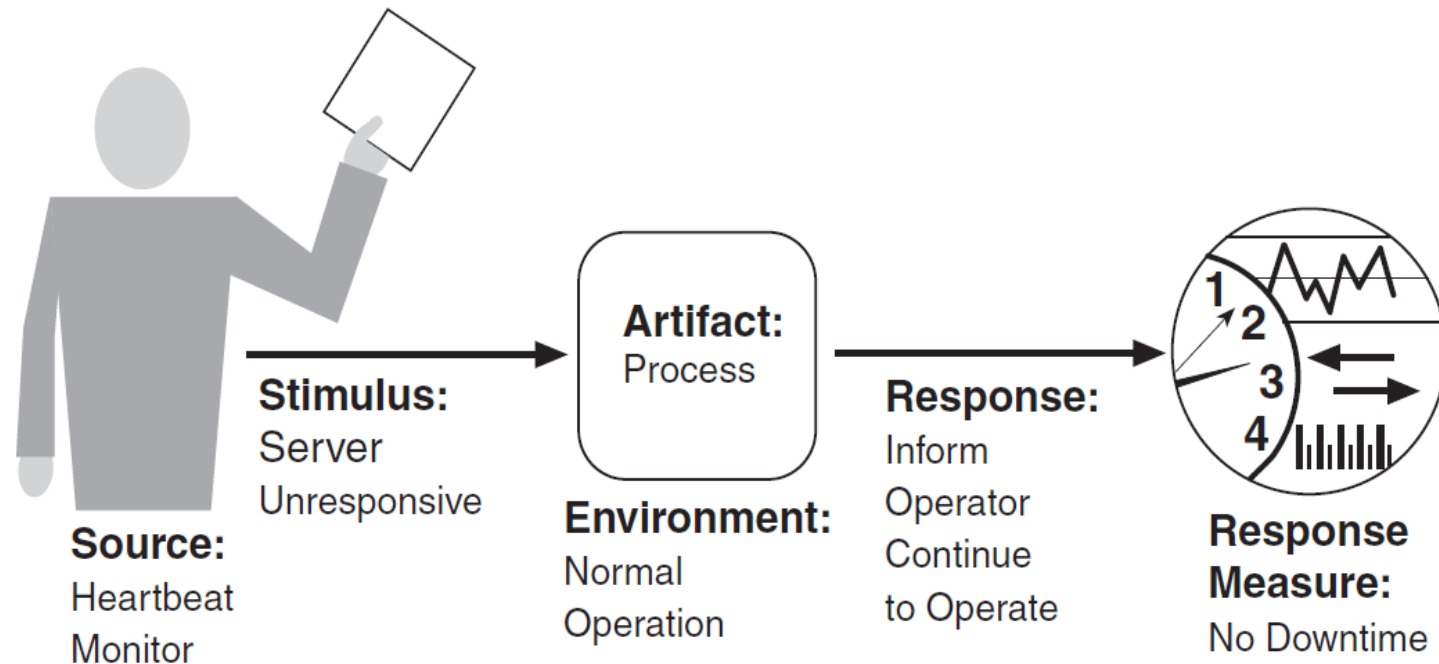
- Availability refers to a **property** of **software** that **it is there** and **ready** to **carry** out its **task when** you **need** it to be.
- This is a **broad perspective** and encompasses **what** is normally **called** **reliability**.
- **Availability** **builds on** **reliability** by **adding** the notion of **recovery** (**repair**).
- **Fundamentally, availability** is about **minimizing service outage time** by **mitigating faults**.

Availability General Scenario

Portion of Scenario	Possible Values
Source	Internal/external: people, hardware, software, physical infrastructure, physical environment
Stimulus	Fault: omission, crash, incorrect timing, incorrect response
Artifact	System's processors, communication channels, persistent storage, processes
Environment	Normal operation, startup, shutdown, repair mode, degraded operation, overloaded operation
Response	Prevent the fault from becoming a failure Detect the fault: • log the fault • notify appropriate entities (people or systems) Recover from the fault • disable source of events causing the fault • be temporarily unavailable while repair is being effected • fix or mask the fault/failure or contain the damage it causes • operate in a degraded mode while repair is being effected
Response Measure	Time or time interval when the system must be available Availability percentage (e.g. 99.999%) Time to detect the fault Time to repair the fault Time or time interval in which system can be in degraded mode Proportion (e.g., 99%) or rate (e.g., up to 100 per second) of a certain class of faults that the system prevents, or handles without failing

Sample Concrete Availability Scenario

- The **heartbeat monitor determines** that the **server** is **nonresponsive** during normal operations. The system informs the operator and continues to operate with no downtime.



Interoperability

- What is Interoperability?
- Interoperability General Scenario

What is Interoperability?

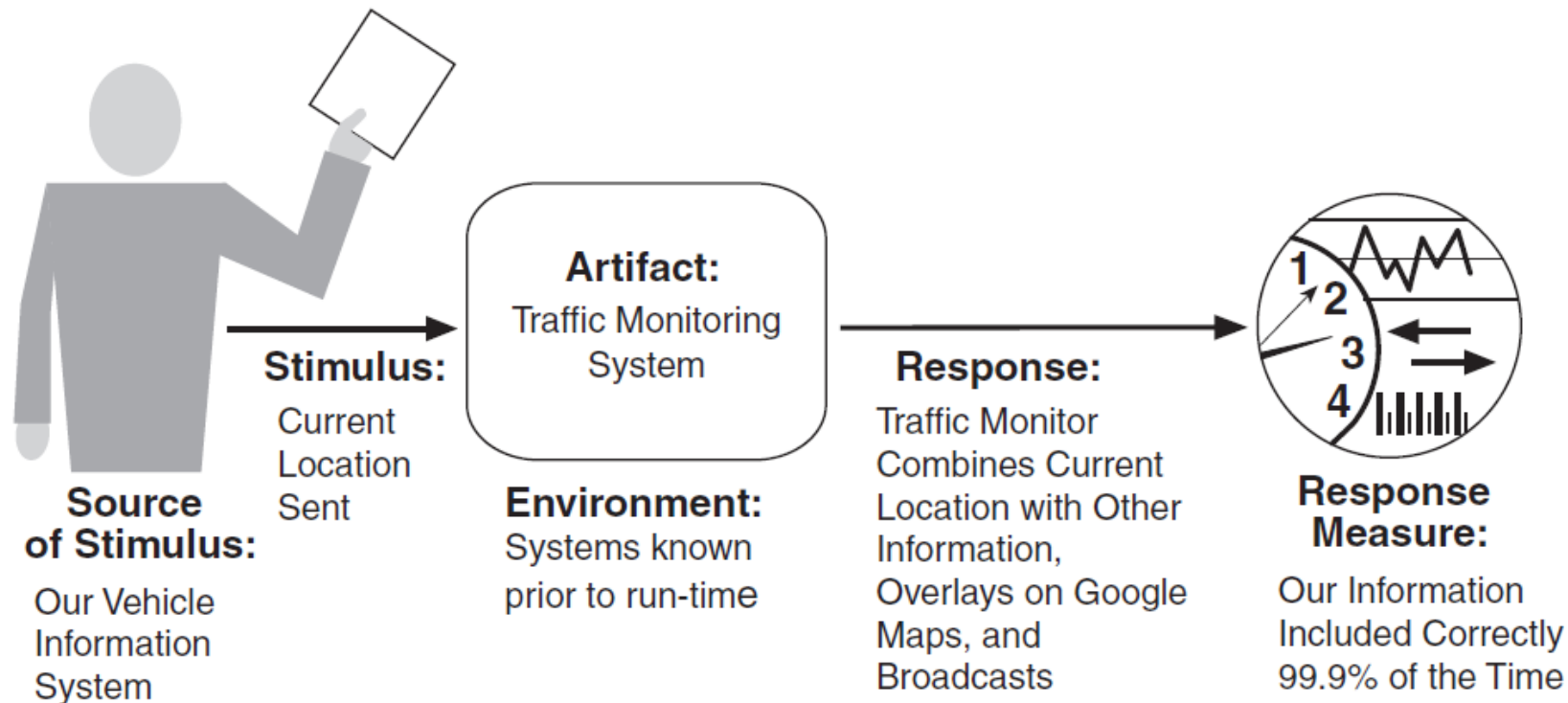
- Interoperability is about the **degree** to which two or more **systems can usefully exchange** meaningful **information**.
- Like all quality attributes, interoperability is not a yes-or-no proposition but has shades of meaning.

Interoperability General Scenario

Portion of Scenario	Possible Values
Source	A system
Stimulus	A request to exchange information among system(s).
Artifact	The systems that wish to interoperate
Environment	System(s) wishing to interoperate are discovered at run time or known prior to run time.
Response	One or more of the following: • the request is (appropriately) rejected and appropriate entities (people or systems) are notified • the request is (appropriately) accepted and information is exchanged successfully • the request is logged by one or more of the involved systems
Response Measure	One or more of the following: • percentage of information exchanges correctly processed • percentage of information exchanges correctly rejected

Sample Concrete Interoperability Scenario

- Our vehicle information system sends our current location to the traffic monitoring system. The traffic monitoring system combines our location with other information, overlays this information on a Google Map, and broadcasts it. Our location information is correctly included with a probability of 99.9%.



Modifiability

- What is Modifiability?
- Modifiability General Scenario

What is Modifiability?

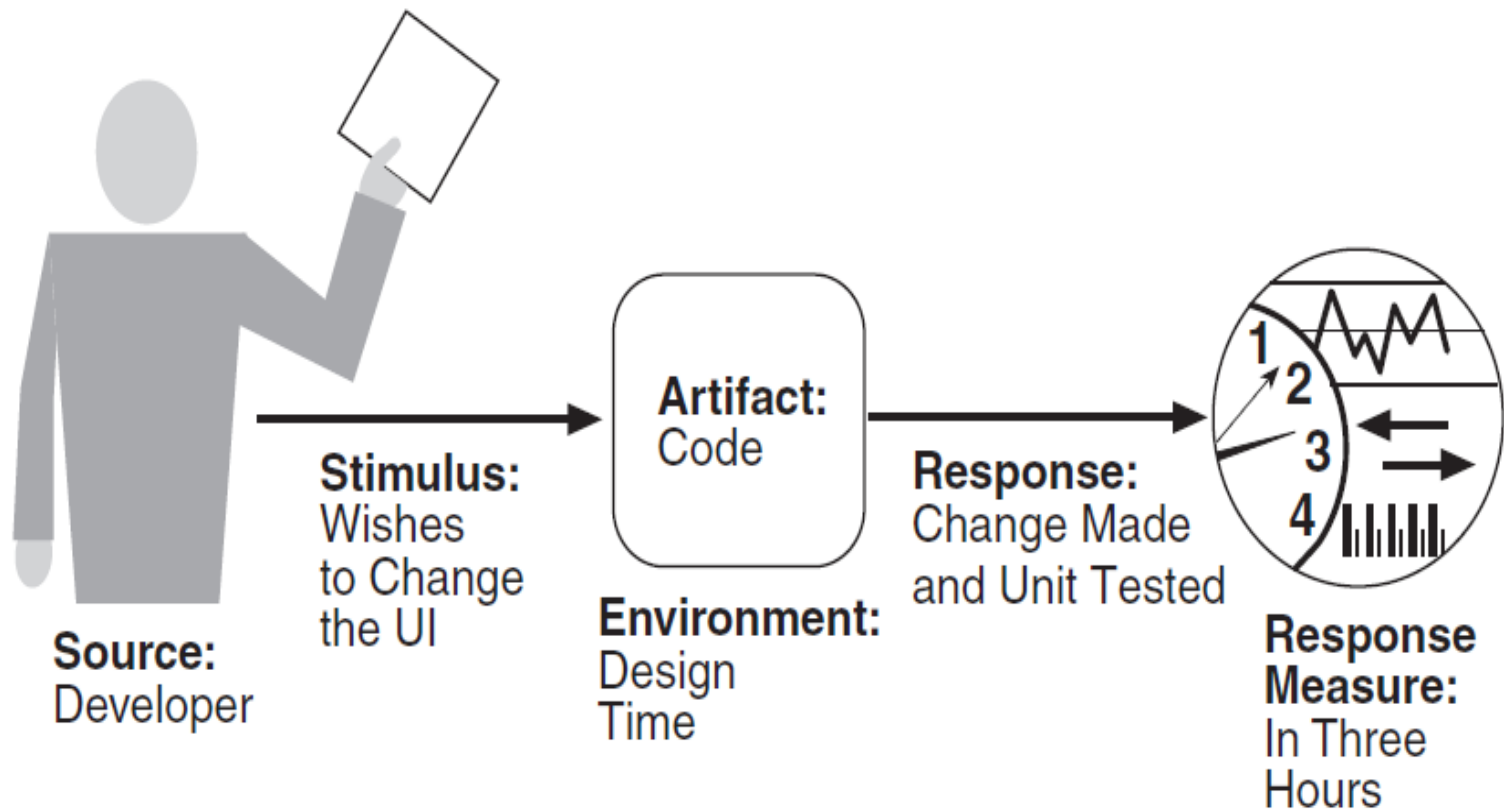
- Modifiability is about **change** and our **interest in** it is in the **cost** and **risk** of making changes.
- To **plan** for **modifiability**, an **architect** has to **consider** three questions:
 - **What can change?**
 - **What is the likelihood of the change?**
 - **When is the change made and who makes it?**

Modifiability General Scenario

Portion of Scenario	Possible Values
Source	End user, developer, system administrator
Stimulus	A directive to add/delete/modify functionality, or change a quality attribute, capacity, or technology
Artifacts	Code, data, interfaces, components, resources, configurations, ...
Environment	Runtime, compile time, build time, initiation time, design time
Response	One or more of the following: • make modification • test modification • deploy modification
Response Measure	Cost in terms of: • number, size, complexity of affected artifacts • effort • calendar time • money (direct outlay or opportunity cost) • extent to which this modification affects other functions or quality attributes • new defects introduced

Sample Concrete Modifiability Scenario

- The developer wishes to change the user interface by modifying the code at design time. The modifications are made with no side effects within three hours.



Performance

- What is Performance?
- Performance General Scenario

What is Performance?

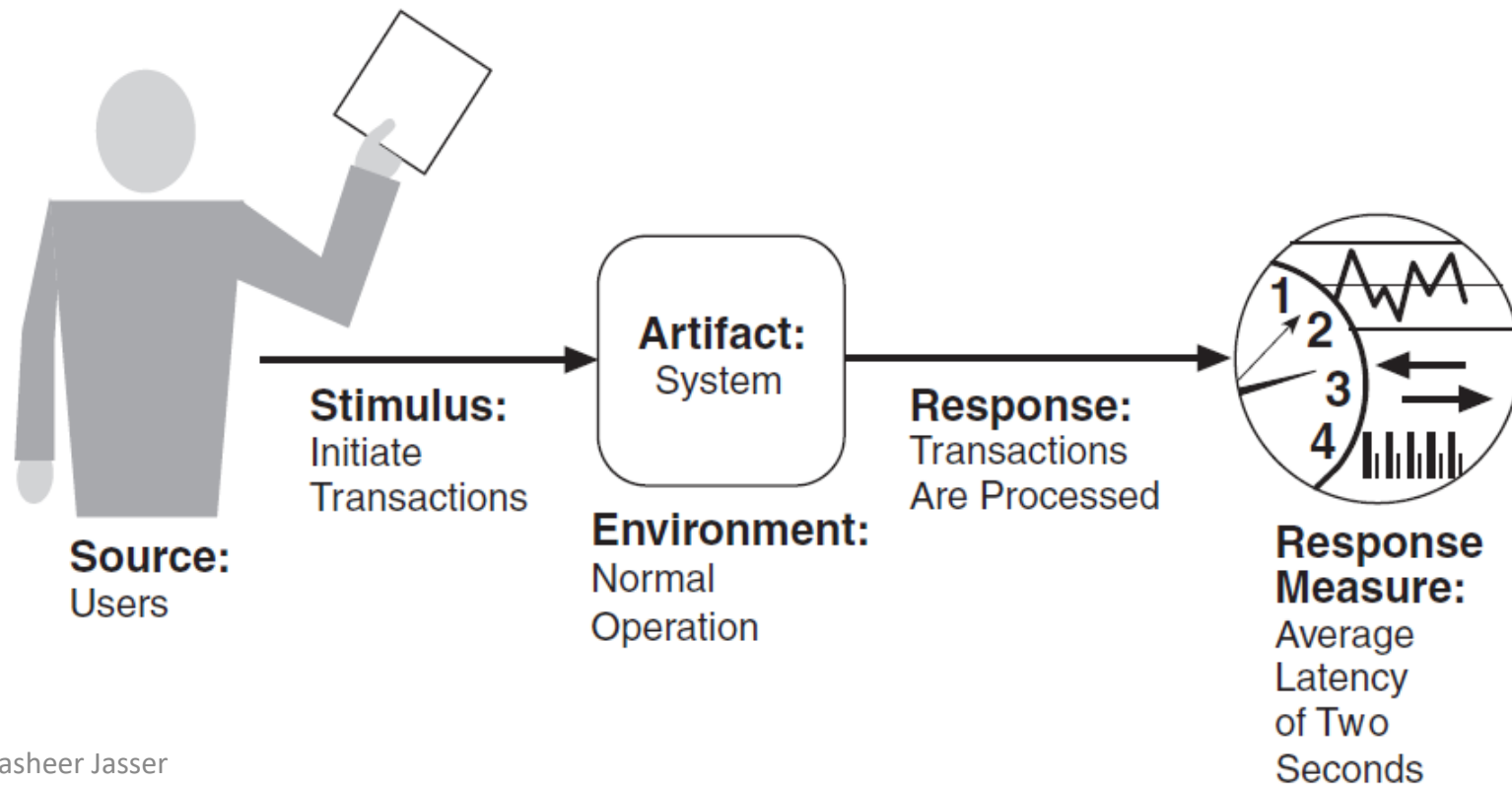
- Performance is about time and the software system's ability to meet timing requirements.
- When events occur – interrupts, messages, requests from users or other systems, or clock events marking the passage of time – the system, or some element of the system, must respond to them in time.
- Characterizing the events that can occur (and when they can occur) and the system or element's time-based response to those events is the essence of discussing performance.

Performance General Scenario

Portion of Scenario	Possible Values
Source	Internal or external to the system
Stimulus	Arrival of a periodic, sporadic, or stochastic event
Artifact	System or one or more components in the system.
Environment	Operational mode: normal, emergency, peak load, overload.
Response	Process events, change level of service
Response Measure	Latency, deadline, throughput, jitter, miss rate

Sample Concrete Performance Scenario

- Users initiate transactions under normal operations. The system processes the transactions with an average latency of two seconds.



Security

- What is Security?
- Security General Scenario

What is Security?

- Security is a **measure** of the **system's ability** to **protect** data and **information** from **unauthorized access** while still providing access to **people** and **systems** that are **authorized**.
- An action taken against a computer system with the intention of doing harm is called an *attack* and can take a number of forms.
- It may be an unauthorized attempt to access data or services or to modify data, or it may be intended to deny services to legitimate users.

What is Security?

- **Security** has **three** main **characteristics**, called **CIA**:
 - **Confidentiality** is the property that data or services are protected from unauthorized access. For example, a hacker cannot access your income tax returns on a government computer.
 - **Integrity** is the property that data or services are not subject to unauthorized manipulation. For example, your grade has not been changed since your instructor assigned it.
 - **Availability** is the property that the system will be available for legitimate use. For example, a denial-of-service attack won't prevent you from ordering a book from an online bookstore.

What is Security?

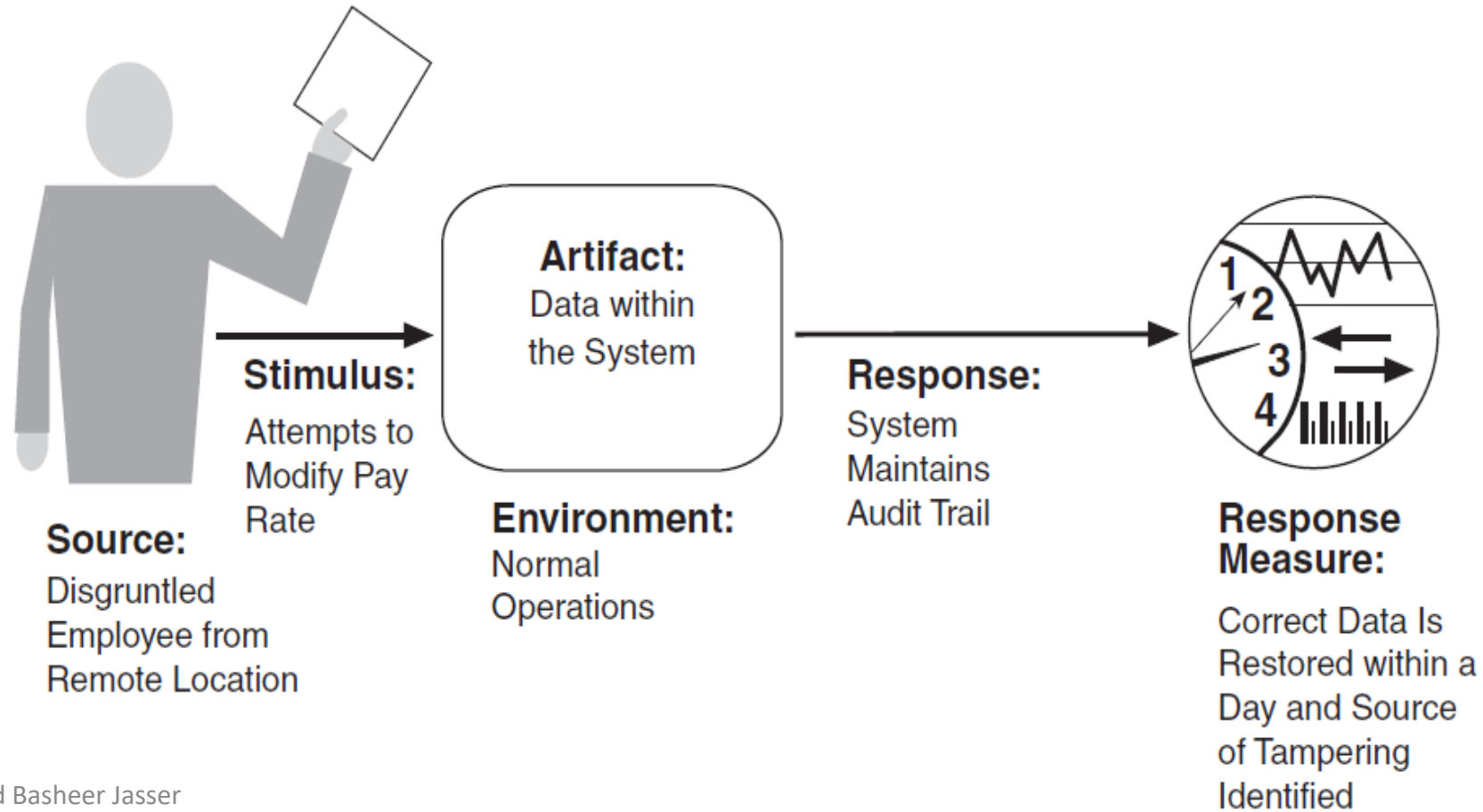
- Other characteristics that support CIA are
 - Authentication verifies the identities of the parties to a transaction and checks if they are truly who they claim to be.
 - For example, when you get an e-mail purporting to come from a bank, authentication guarantees that it actually comes from the bank.
 - Nonrepudiation guarantees that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message. For example, you cannot deny ordering something from the Internet, or the merchant cannot disclaim getting your order.
 - Authorization grants a user the privileges to perform a task.
 - For example, an online banking system authorizes a legitimate user to access his account.

Security General Scenario

Portion of Scenario	Possible Values
Source	Human or another system which may have been previously identified (either correctly or incorrectly) or may be currently unknown. A human attacker may be from outside the organization or from inside the organization.
Stimulus	Unauthorized attempt is made to display data, change or delete data, access system services, change the system's behavior, or reduce availability.
Artifact	System services; data within the system; a component or resources of the system; data produced or consumed by the system
Environment	The system is either online or offline, connected to or disconnected from a network, behind a firewall or open to a network, fully operational, partially operational, or not operational
Response	Transactions are carried out in a fashion such that • data or services are protected from unauthorized access; • data or services are not being manipulated without authorization; • parties to a transaction are identified with assurance; • the parties to the transaction cannot repudiate their involvements; • the data, resources, and system services will be available for legitimate use. The system tracks activities within it by • recording access or modification, • recording attempts to access data, resources or services, • notifying appropriate entities (people or systems) when an apparent attack is occurring.
Response Measure	One or more of the following • how much of a system is compromised when a particular component or data value is compromised, • how much time passed before an attack was detected, • how many attacks were resisted, • how long does it take to recover from a successful attack, • how much data is vulnerable to a particular attack

Sample Concrete Security Scenario

- A disgruntled **employee** from a **remote location** **attempts** to **modify** the **pay rate** table during normal operations. The system maintains an audit trail and the correct data is restored within a day.



Testability

- What is Testability?
- Testability General Scenario

What is Testability?

- Software testability refers to the **ease** with which **software** can be **made** to **demonstrate** its **faults through** (typically execution-based) **testing**.
- Specifically, testability refers to the probability, assuming that the software has at least one fault, that it will fail on its *next* test execution.
- **If** a **fault** is **present** in a **system**, then **we want it** to **fail during testing** as **quickly** as possible.

What is Testability?

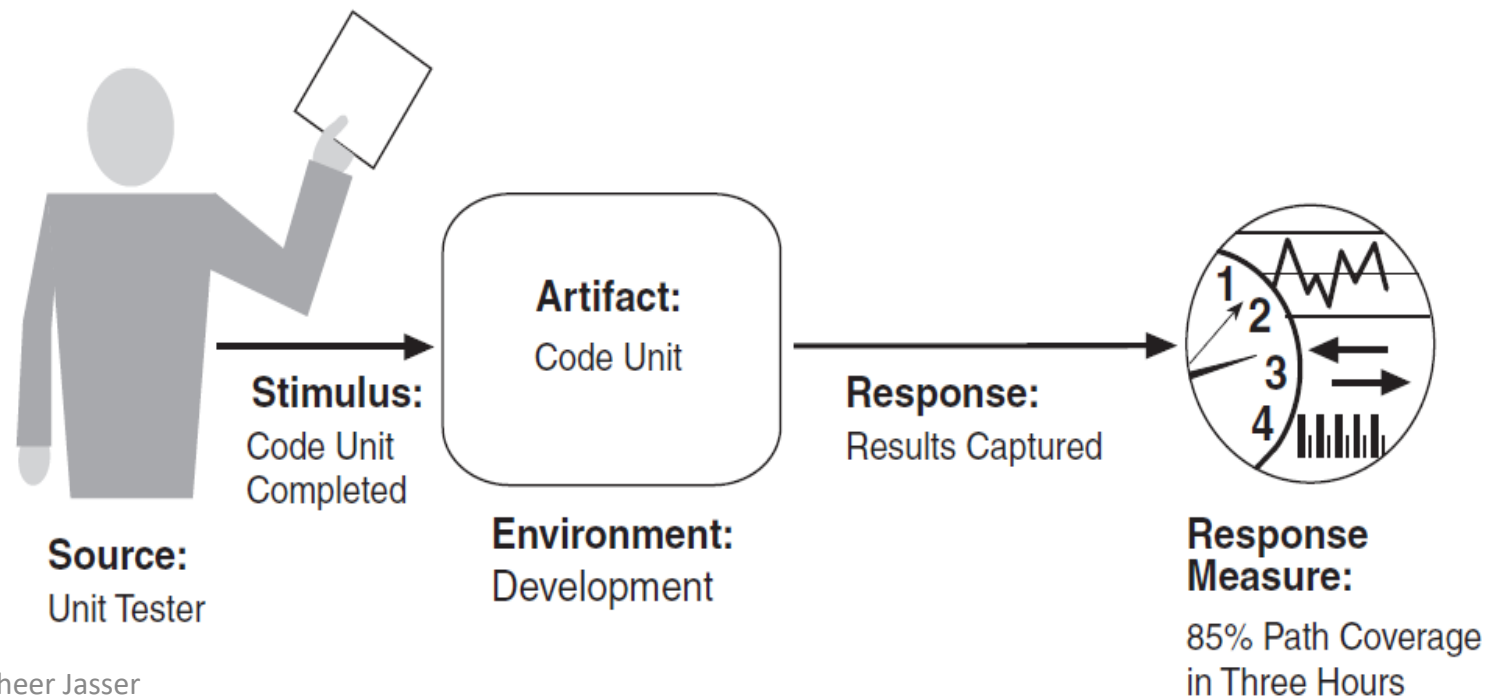
- For a *system* to be properly *testable*, it must be *possible* to *control* each *component's inputs* (and *possibly* manipulate its *internal state*) and then to *observe* its *outputs* (and *possibly* its *internal state*).

Testability General Scenario

Portion of Scenario	Possible Values
Source	Unit testers, integration testers, system testers, acceptance testers, end users, either running tests manually or using automated testing tools
Stimulus	A set of tests are executed due to the completion of a coding increment such as a class, layer or service; the completed integration of a subsystem; the complete implementation of the system; or the delivery of the system to the customer.
Environment	Design time, development time, compile time, integration time, deployment time, run time
Artifacts	The portion of the system being tested
Response	One or more of the following: execute test suite and capture results; capture activity that resulted in the fault; control and monitor the state of the system
Response Measure	One or more of the following: effort to find a fault or class of faults, effort to achieve a given percentage of state space coverage; probability of fault being revealed by the next test; time to perform tests; effort to detect faults; length of longest dependency chain in test; length of time to prepare test environment; reduction in risk exposure ($\text{size}(\text{loss}) * \text{prob}(\text{loss})$)

Sample Concrete Testability Scenario

- The unit tester completes a code unit during development and performs a test sequence whose results are captured and that gives 85% path coverage within 3 hours of testing.



Usability

- What is Usability?
- Usability General Scenario

What is Usability?

- Usability is concerned with **how easy** it is for the **user** to **accomplish** a **desired task** and the kind of **user support** the **system provides**.
- Over the years, a focus on usability has shown itself to be one of the cheapest and easiest ways to improve a system's quality (or, more precisely, the user's *perception* of quality).

What is Usability?

- Usability comprises the following areas:
 - Learning system features.
 - Using a system efficiently.
 - Minimizing the impact of errors.
 - Adapting the system to user needs.
 - Increasing confidence and satisfaction.

Usability General Scenario

Portion of Scenario	Possible Values
Source	End user, possibly in a specialized role
Stimulus	End user tries to use a system efficiently, learn to use the system, minimize the impact of errors, adapt the system, or configure the system
Environment	Runtime or configuration time
Artifacts	System or the specific portion of the system with which the user is interacting.
Response	The system should either provide the user with the features needed or anticipate the user's needs.
Response Measure	One or more of the following: task time, number of errors, number of tasks accomplished, user satisfaction, gain of user knowledge, ratio of successful operations to total operations, or amount of time or data lost when an error occurs.

Sample Concrete Usability Scenario

- The user downloads a new application and is using it productively after two minutes of experimentation.

